A conditional stationary-headway interpretation of H1's capacity curve

VSL task-definition review: Step 3 supporting note

14 October 2026

Status. Source-backed stationary calculation and original algebraic correspondence. No vehicle trajectory, traffic experiment, parameter fit or new H1/T1 law is supplied. The approved model is unchanged. This note supports the plain-language reference-qualification report, not an empirical capacity or gradient claim.

1. Why speed can change usable lane density

The official SUMO Road Capacity documentation gives the minimum gross front-to-front spacing for homogeneous Krauss vehicles travelling at the same constant speed. Let vehicle length be l_{veh} in metres, standstill gap $g_{\min}$ in metres and desired following time $\tau_d > 0$ in seconds. Write $d = l_{\text{veh}} + g_{\min} > 0$. At actual speed $s > 0$, the documented stationary relation and its ideal flow consequence are

$$D(s) = d + \tau_d s, \qquad [D] = \text{m}, \tag{1}$$

$$k_{\text{ideal}}(s) = \frac{1}{d + \tau_d s}, \qquad [k_{\text{ideal}}] = \text{veh/m}, \tag{2}$$

$$Q_{\text{ideal}}(s) = \frac{s}{d + \tau_d s}, \qquad [Q_{\text{ideal}}] = \text{veh/s}. \tag{3}$$

The documentation cautions that these conditions are rarely achieved exactly in a simulation. Desired-speed differences, random slowdowns and nonminimum gaps are among its stated reasons. This is an ideal stationary relation, not the realised capacity of the proposed lane drop.

2. Algebraic correspondence with the accepted H1 curve

H1 uses triangular-diagram critical density and capacity

$$k_c(v) = \frac{w k_j}{v + w}, \qquad C(v) = v k_c(v). \tag{4}$$

For the sole purpose of interpreting the stationary curve, set

$$k_j = \frac{1}{d}, \qquad w = \frac{d}{\tau_d}. \tag{5}$$

Substitution gives

$$k_c(v) = \frac{1}{d + \tau_d v}, \qquad C(v) = \frac{v}{d + \tau_d v}. \tag{6}$$

Thus the two curve shapes coincide if the stationary actual speed s equals the predictor's advice v. This equality of shapes is an inference from the documented headway law. It does not establish equality of the two dynamic traffic models or prescribe reference parameters.

3. The traffic meaning and its limits

The directions of the stationary changes are

$$\frac{dk_{\text{ideal}}}{ds} = -\frac{\tau_d}{(d+\tau_d s)^2} < 0, \qquad \frac{dQ_{\text{ideal}}}{ds} = \frac{d}{(d+\tau_d s)^2} > 0. \tag{7}$$

Slower travel permits more closely spaced vehicles per metre at the same desired following time, while ideal vehicles-per-second throughput falls. This gives a physical interpretation to density headroom in H1 without claiming that lower speed increases every kind of service. In H1's progress coordinate, the explicit speed factor in the preparation law cancels: the remaining preparation effect comes from headroom and the model's specified interaction, not additional seconds by themselves.

The following conclusions do *not* follow from the correspondence:

1. A native lane-drop interaction obeys H1's receiving allocator, preparation opportunity scale or forced-service work law.
2. $w = d/\tau_d$ is a measured transient backward-wave speed in the chosen microscopic model. The equality identifies a stationary curve only.
3. Native actual speed equals the permitted advice during acceleration, braking, lane changes or queueing.
4. A positive analytical work-factor range is traffic-calibrated, or an interior optimal advice exists.

For example, the earlier analytical family chose $w = 2u$. Under this optional stationary interpretation it would imply $u = d/(2\tau_d)$. Its dimensionless construction must therefore not be presented as already calibrated freeway speeds. The general approved family is broader; no numerical mapping is selected here.

4. Why a mixed departure trace may not identify the forced server

On intervals where the accepted fluid server is binding, let u_A be continuing departure flow and g_E ending departure flow, both in vehicle-equivalents per second. Its work account has the diagnostic form

$$u_A + \alpha g_E = R. \tag{8}$$

One repeated mixed-flow value supplies only one relation for the two unknowns α and R. At least differing flow observations with a nondegenerate design, correct binding-interval classification and a valid common-law fit are needed to separate them; two different g_E values are necessary for rank two in this simple linear diagnostic. Native stopped vehicles do not by themselves certify the fluid server's binding condition. If the law fails across independently prepared development traces, or parameters remain unidentifiable, retain that failure or the admissible bounds. Do not select α from the positive-action range or fit held-out optima. This is an identifiability observation, not a fitted native model.

Primary source inspected, 14 October 2026: SUMO documentation, Road Capacity, sections Computing Headways and Further headway effects. All correspondence, derivatives and identifiability reasoning above are the review's deductions under their stated conditions.